

jQuery Exercises

Exercise 1: Adding jQuery to a Page

index.html - jQuery included via CDN (commented) and via a locally downloaded copy, with a Hello World paragraph logged to the console:

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Exercise 1 - jQuery Inclusion</title>
<!-- Option 1: Using a CDN link -->
<!-- <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script> -->

<!-- Option 2: Using a locally downloaded copy -->
<script src="jquery.min.js"></script>
</head>
<body>
  <p id="msg">Hello World!</p>

  <script>
    $(document).ready(function () {
      console.log($('#msg').text());
    });
  </script>
</body>
</html>
```

Page output:

Hello World!

Browser console output:

Hello World!

Exercise 2: Selecting Elements

```
<h1 id="heading">Original Heading</h1>
<p class="text">This is paragraph one.</p>
<p class="text" id="para2">This is paragraph two.</p>
<button id="changeBtn">Change Text & Hide Paragraph</button>

<script>
  $(document).ready(function () {
    $('#changeBtn').click(function () {
      $('#heading').text('Heading Changed by jQuery!');
      $('#para2').hide();
    });
  });
</script>
```

Before clicking the button:

Original Heading

This is paragraph one.

This is paragraph two.

Change Text & Hide Paragraph

After clicking the button - heading changed and second paragraph hidden:

Heading Changed by jQuery!

This is paragraph one.

Change Text & Hide Paragraph

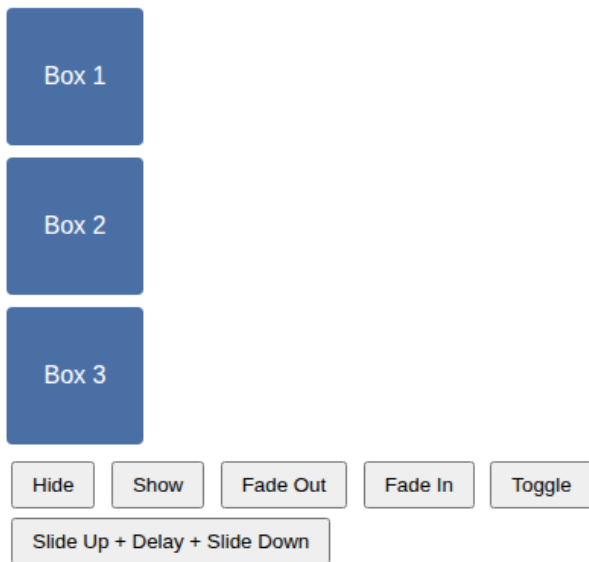
Exercise 3: Use of Common Methods

```
<div class="box" id="box1">Box 1</div>
<div class="box" id="box2">Box 2</div>
<div class="box" id="box3">Box 3</div>

<button id="hideBtn">Hide</button>
<button id="showBtn">Show</button>
<button id="fadeOutBtn">Fade Out</button>
<button id="fadeInBtn">Fade In</button>
<button id="toggleBtn">Toggle</button>
<button id="chainBtn">Slide Up + Delay + Slide Down</button>

<script>
  $(document).ready(function () {
    $('#hideBtn').click(function () { $('.box').hide(); });
    $('#showBtn').click(function () { $('.box').show(); });
    $('#fadeOutBtn').click(function () { $('.box').fadeOut(); });
    $('#fadeInBtn').click(function () { $('.box').fadeIn(); });
    $('#toggleBtn').click(function () { $('.box').toggle(); });
    $('#chainBtn').click(function () {
      $('#box1').slideUp().delay(1000).slideDown();
    });
  });
</script>
```

Initial state - all 3 boxes visible:



After clicking Hide - all boxes hidden:



After Show, then Toggle - boxes hidden again since toggle flips visibility:



Exercise 4: Creating and Modifying Elements

```
<input type="text" id="itemInput" placeholder="Enter item">
<button id="addBtn">Add</button>
<button id="removeAllBtn">Remove All</button>
<ul id="itemList"></ul>

<script>
  $(document).ready(function () {
    $('#addBtn').click(function () {
      var value = $('#itemInput').val();
      if (value.trim() !== '') {
        $('#itemList').append('<li>' + value + '</li>');
        $('#itemInput').val('');
      }
    });
  });
</script>
```

```
$('#removeAllBtn').click(function () {  
    $('#itemList').empty();  
});  
});  
</script>
```

Before adding any item:

After typing values and clicking Add twice:

- Buy groceries
- Finish assignment

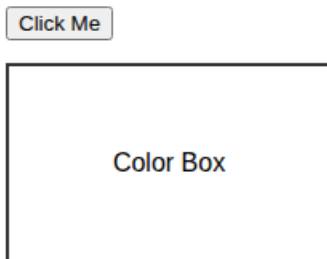
After clicking Remove All:

Exercise 5: Handling User Interactions

```
<button id="colorBtn">Click Me</button>  
<div id="colorBox">Color Box</div>  
  
<script>  
    $(document).ready(function () {  
        $('#colorBtn').click(function () {
```

```
$('#colorBox').css('background-color', 'red');  
});  
  
$('#colorBox').dblclick(function () {  
    $(this).css('background-color', 'white');  
});  
});  
</script>
```

Before clicking the button:



After clicking the button - box turns red:



After double-clicking the box - box turns back to white:



Exercise 6: Mouse and Keyboard Events

```
<div id="textBox">Hover, click or double-click me</div>
<input type="text" id="keyInput" placeholder="Type something...">
<p id="keyOutput"></p>

<script>
  $(document).ready(function () {
    $('#textBox').click(function () {
      $(this).css('background-color', '#cce5ff');
    });
    $('#textBox').dblclick(function () {
      $(this).text('Double-clicked!');
    });
    $('#textBox').mouseenter(function () {
      $(this).css('background-color', '#d4edda');
    });
    $('#textBox').mouseleave(function () {
      $(this).css('background-color', '#ffffff');
    });
    $('#keyInput').keypress(function () {
      $('#keyOutput').text('Key pressed! Current value: ' + $(this).val());
    });
  });
</script>
```

Before any interaction:

Hover, click or double-click me

Type something...

On mouseenter - background turns light green:

Hover, click or double-click me

Type something...

After typing in the input field - keypress event updates the output text:

Hover, click or double-click me

Hello

Key pressed! Current value: Hell